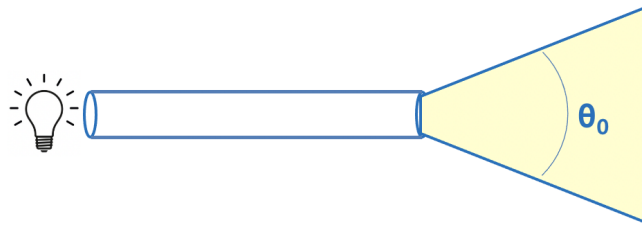


Y2 Electromagnetism 2

Non-assessed problem sheet 5 (weeks 9–11)

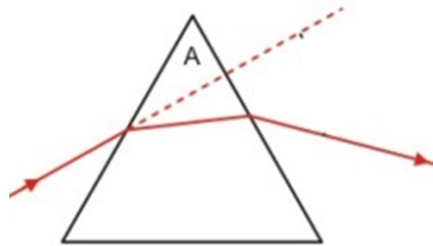
EM waves in dielectrics, conductors and plasma

1) Consider a long thin cylindrical lightguide made of glass with a refractive index n . A point light source is located on the axis of symmetry of the lightguide close to the entrance. Find the opening angle θ_0 of the light cone at the exit of the lightguide. Plot the dependence $\theta_0 = f(n)$, and establish its properties.



2) Consider a plane electromagnetic wave traversing a boundary from a dielectric medium A into a dielectric medium B. The angle of incidence is equal to the Brewster's angle α_B , and the angle of refraction is α . Prove that for a wave traversing the boundary from the medium B into the medium A, the angle α is the Brewster's angle.

3) Consider monochromatic light travelling in vacuum and traversing a right triangular prism with an apex angle A made of a dielectric material. Which conditions must be satisfied for all of the incoming energy to be transmitted, and none to be reflected? Find the refractive index of the prism for $A = 60^\circ$ required for the absence of reflection.



4) The *degree of polarisation* of reflected and refracted natural light at a boundary of two dielectric media is defined in terms of the reflectances (R) and transmittances (T) of the s -polarised ($\perp$) and p -polarised ($\parallel$) light as follows:

$$\Delta_R = \frac{R_\perp - R_\parallel}{R_\perp + R_\parallel}, \quad \Delta_T = \frac{T_\perp - T_\parallel}{T_\perp + T_\parallel}.$$

Consider a ray of light incident on a vacuum/glass boundary at the Brewster's angle. The refractive index of glass is n . Using the Fresnel's equations and Brewster's angle properties, compute Δ_R and Δ_T as functions of n . Give a numerical answer for $n = 1.5$.

5) Show that the conductivity of plasma is purely imaginary: $\sigma = -in_e e^2 / m_e \omega$.